ប្រមាណវិធីលើរ៉ាឌីកាល់សន្ទស្សន៍ 2 និង 3

★ រំលឹកទ្រឹស្តី និងរូបមន្តគន្លឹះសំខាន់ៗ

1. **និយមន័យ ៖** បើ $a>0,\ x^2=a \implies x=\pm\sqrt{a}$ | បើ $x^3=a \implies x=\sqrt[3]{a}$ ដែល $a\in\mathbb{R}$ ។
2. **លក្ខណៈសំខាន់** $(a,b>0,b\neq 0)$ **៖** $\sqrt{a}\cdot\sqrt{b}=\sqrt{ab}$, $\sqrt[3]{a}\cdot\sqrt[3]{b}=\sqrt[3]{ab}$, $\dfrac{\sqrt{a}}{\sqrt{b}}=\sqrt{\dfrac{a}{b}}$, $\dfrac{\sqrt[3]{a}}{\sqrt[3]{b}}=\sqrt[3]{\dfrac{a}{b}}$ ។
3. **វិធីបូក-ដករ៉ាឌីកាល់ ៖** ត្រូវបម្លែងឱ្យសន្ទស្សន៍ និងរ៉ាឌីកង់ដូចគ្នា រួចបូកឬដកតែមេគុណ ។
4. **បំបាត់រ៉ាឌីកាល់ពីភាគបែង ៖**
   - ➢ $(a-b)(a+b)=a^2-b^2$ និង $a^3\pm b^3=(a\pm b)(a^2\mp ab+b^2)$
   - ➢ *ភាគបែងជាឬសការេ ៖* $\dfrac{1}{\sqrt{7}-1}=\dfrac{\sqrt{7}+1}{(\sqrt{7}-1)(\sqrt{7}+1)}=\dfrac{\sqrt{7}+1}{7-1}=\dfrac{\sqrt{7}+1}{6}$
   - ➢ *ភាគបែងជាឬសគូប ៖* $\dfrac{1}{\sqrt[3]{2}+1}=\dfrac{\sqrt[3]{4}-\sqrt[3]{2}+1}{(\sqrt[3]{2}+1)(\sqrt[3]{4}-\sqrt[3]{2}+1)}=\dfrac{\sqrt[3]{4}-\sqrt[3]{2}+1}{3}$

I គណនាឫសការេ និងឫសគូប

01 គណនា ៖

➥ **តំរូ ៖** ក. $\sqrt{9}=3$ ឃ. $\sqrt[3]{8}=2$ ត. $\sqrt{\frac{9}{16}}=\frac{3}{4}$ ផ. $\sqrt[3]{\frac{1}{8}}=\frac{1}{2}$

ក. $\sqrt{9}$ ខ. $\sqrt{16}$ គ. $\sqrt{36}$ ឃ. $-\sqrt{64}$
ង. $-\sqrt{100}$ ច. $\sqrt{121}$ ឆ. $-\sqrt{144}$ ជ. $\sqrt{625}$
ឈ. $\sqrt[3]{8}$ ញ. $\sqrt[3]{-8}$ ដ. $-\sqrt[3]{27}$
ឋ. $-\sqrt[3]{64}$ ឌ. $\sqrt[3]{125}$ ឍ. $\sqrt[3]{216}$
ណ. $\sqrt[3]{1000}$ ត. $\sqrt{\frac{9}{16}}$ ថ. $\sqrt{\frac{49}{9}}$
ទ. $-\sqrt{\frac{81}{4}}$ ធ. $\sqrt{\frac{169}{49}}$ ន. $\sqrt{\frac{196}{25}}$
ប. $-\sqrt{\frac{400}{225}}$ ផ. $\sqrt[3]{\frac{1}{8}}$ ព. $\sqrt[3]{\frac{8}{27}}$
ភ. $\sqrt[3]{\frac{64}{125}}$ ម. $\sqrt[3]{\frac{512}{343}}$

02 សម្រួលរ៉ាឌីកាល់ ៖

➥ **តំរូ ៖** ខ. $(\sqrt[3]{-8})^3=-8$ ឃ. $\sqrt[3]{8^2}=2^2=4$ ទ. $\sqrt{x^2y^4}=|x|y^2$

ក. $\sqrt[3]{1^5}$ ខ. $(\sqrt[3]{-8})^3$ គ. $\sqrt[3]{(-27)^3}$
ឃ. $\sqrt[3]{8^2}$ ង. $\sqrt[3]{(-27)^2}$ ច. $\sqrt[3]{64^2}$
ឆ. $\sqrt{16^3}$ ជ. $\sqrt{25^3}$ ឈ. $\sqrt{64^3}$
ញ. $-\sqrt[3]{36^2}$ ដ. $\sqrt{y^2}$ ឋ. $\sqrt[3]{x^3}$
ឌ. $\sqrt{x^4}$ ឍ. $\sqrt[3]{y^6}$ ណ. $\sqrt{y^6}$
ត. $\sqrt{25a^2}$ ថ. $\sqrt[3]{-8b^3}$ ទ. $\sqrt{x^2y^4}$
ធ. $\sqrt{36a^4b^6}$ ន. $\sqrt{\frac{16}{x^2}}$ ប. $\sqrt[3]{\frac{27}{y^3}}$
ផ. $\sqrt{\frac{100}{n^4}}$ ព. $\sqrt{\frac{49a^2}{81b^4}}$

03 សម្រួលកន្សោម ៖

➥ **តំរូ ៖** ក. $\sqrt{(2x)^2}=2x$ ខ. $\sqrt[3]{(-5y)^3}=-5y$ ង. $\sqrt{(4b+3)^2}=|4b+3|$

ក. $\sqrt{(2x)^2}$ ខ. $\sqrt[3]{(-5y)^3}$
គ. $\sqrt{(4-a)^2}$ ឃ. $\sqrt[3]{(x+3)^3}$
ង. $\sqrt{16b^2+24b+9}$
ច. $\sqrt{9x^2-30x+25}$
ឆ. $\sqrt{4m^2-20mn+25n^2}$
ជ. $\sqrt{49x^2-112xy+64y^2}$

04 សម្រួលកន្សោមប្រភាគ ៖

➥ **តំរូ ៖** $A=\sqrt{\frac{b^2}{(b-7)^2}}=\left|\frac{b}{b-7}\right|$

$A=\sqrt{\dfrac{b^2}{b^2-14b+49}}$ $B=\sqrt{\dfrac{49x^2-56x+16}{36x^2}}$
$C=\sqrt{\dfrac{a^2+16ab+64b^2}{a^2+10ab+25b^2}}$ $D=\sqrt{\dfrac{25b^2+10ab+a^2}{16b^2+24ab+9a^2}}$

05 សម្រួលកន្សោមរ៉ាឌីកាល់ឌុប ៖

➥ **តំរូ ៖** $A=\sqrt{22-12\sqrt{2}}=3\sqrt{2}-2$ $B=\sqrt{(2-\sqrt{3})^2}=2-\sqrt{3}$

$A=\sqrt{22-\sqrt{288}}$ $B=\sqrt{7-2\sqrt{12}}$
$C=\sqrt{9+\sqrt{32}}$ $D=\sqrt{13-\sqrt{48}}$
$E=\sqrt{3+\sqrt{5}}$ $F=\sqrt{6+\sqrt{11}}$

06 ប្រៀបធៀបចំនួន $\sqrt{2}$ និង $\sqrt[3]{3}$ ៖

➥ **តំរូ ៖** លើកជាស្វ័យគុណ 6 ៖ $(\sqrt{2})^6=2^3=8<(\sqrt[3]{3})^6=3^2=9 \implies \sqrt{2}<\sqrt[3]{3}$

II បញ្ចេញ និងបញ្ចូលមួយចំនួនពីរ៉ាឌីកាល់

07 បញ្ចេញមួយចំនួនពីរ៉ាឌីកាល់ ៖

➥ **តំរូ ៖** ក. $\sqrt{18}=3\sqrt{2}$ ច. $\sqrt[3]{40}=2\sqrt[3]{5}$ ឋ. $\sqrt{36a^2b^3}=6|a|b\sqrt{b}$

ក. $\sqrt{18}$ ខ. $-\sqrt{48}$ គ. $\sqrt{75}$ ឃ. $\sqrt{\frac{30}{49}}$

ង. $\sqrt{\frac{10}{121}}$ ច. $\sqrt[3]{40}$ ឆ. $\sqrt[3]{54}$ ជ. $-\sqrt[3]{128}$

ឈ. $\sqrt[3]{192}$ ញ. $\sqrt[3]{\frac{3m}{8n^3}}$

ដ. $\sqrt[3]{16a^5}$ ឋ. $\sqrt{36a^2b^3}$

ឌ. $\sqrt{27a^4b^3}$ ឍ. $\sqrt{72x^5y^2}$

ណ. $\sqrt{80m^4n^3}$ ត. $-\sqrt{112a^3b^4}$

ថ. $\sqrt{64x^2y^3}$ ទ. $\sqrt[3]{16m^3n^3}$

ធ. $\sqrt[3]{-54x^4b^3}$ ន. $-\sqrt[3]{128a^5y^3}$

08 បញ្ចូលមួយចំនួនក្នុងរ៉ាឌីកាល់ $(x,y,m,n,p>0)$ ៖

➥ **តំរូ ៖** ក. $5\sqrt{6}=\sqrt{5^2\times 6}=\sqrt{150}$ ឃ. $2\sqrt[3]{5}=\sqrt[3]{2^3\times 5}=\sqrt[3]{40}$

ក. $5\sqrt{6}$ ខ. $2m\sqrt{m}$ គ. $\frac{\sqrt{23}}{y^3}$

ឃ. $2\sqrt[3]{5}$ ង. $2x\sqrt[3]{4}$ ច. $\sqrt[3]{\frac{3m}{2n}}$

III ប្រមាណវិធីលើរ៉ាឌីកាល់ (បូក ដក គុណ)

09 គណនា ៖

➥ **តំរូ ៖** ក. $(3-4+5-3)\sqrt{2}=\sqrt{2}$ ង. $3\sqrt[3]{2}-5\sqrt[3]{3}$

ក. $3\sqrt{2}-4\sqrt{2}+5\sqrt{2}-3\sqrt{2}$

ខ. $5\sqrt{2}-3\sqrt{3}-6\sqrt{2}+5\sqrt{3}$

គ. $3\sqrt{15}-4\sqrt{3}-3\sqrt{15}+6\sqrt{3}$

ឃ. $4\sqrt{3}-2\sqrt{17}+3\sqrt{17}-3\sqrt{3}-2\sqrt{3}$

ង. $2\sqrt[3]{2}-8\sqrt[3]{3}+\sqrt[3]{2}+3\sqrt[3]{3}$

ច. $8\sqrt[3]{2}-3\sqrt[3]{3}-5\sqrt[3]{2}+2\sqrt[3]{3}$

10 គណនា ៖

➥ **តំរូ ៖** ក. $\frac{2}{3}(3\sqrt{3})-\frac{3}{4}(4\sqrt{3})=-\sqrt{3}$ ឃ. $20\sqrt[3]{2}-15\sqrt[3]{2}=5\sqrt[3]{2}$

ក. $\frac{2}{3}\sqrt{27}-\frac{3}{4}\sqrt{48}$ ខ. $\frac{1}{4}\sqrt{288}-\frac{1}{6}\sqrt{72}$

គ. $\frac{3}{5}\sqrt{75}-\frac{2}{3}\sqrt{27}$ ឃ. $5\sqrt[3]{128}-3\sqrt[3]{250}$

ង. $3\sqrt[3]{81}-\frac{1}{2}\sqrt[3]{192}$ ច. $4\sqrt[3]{54}-3\sqrt[3]{128}$

11 គណនា ៖

➥ **តំរូ ៖** ក. $4\sqrt{2}-21\sqrt{2}-20\sqrt{2}=-37\sqrt{2}$ ង. $5\sqrt[3]{2}$

ក. $2\sqrt{8}-3\sqrt{98}-2\sqrt{200}$

ខ. $-3\sqrt{50}-\sqrt{32}+5\sqrt{200}$

គ. $3\sqrt{175}-2\sqrt{28}+3\sqrt{63}-\sqrt{112}$

ឃ. $\sqrt{108}-2\sqrt{27}-\sqrt{40}-5\sqrt{160}$

ង. $2\sqrt[3]{16}+3\sqrt[3]{54}-2\sqrt[3]{128}$

ច. $3\sqrt[3]{81}+\frac{1}{2}\sqrt[3]{128}-3\sqrt[3]{192}+4\sqrt[3]{54}$

ឆ. $4\sqrt[3]{54}-6\sqrt[3]{81}-4\sqrt[3]{16}+3\sqrt[3]{24}$

ជ. $-2\sqrt[3]{40}-3\sqrt[3]{135}+5\sqrt[3]{320}+8\sqrt[3]{5}$

12 គណនា ៖

➥ **តំរូ ៖** ក. $-4\sqrt{12}+2\sqrt{18}-15\sqrt{32}+5\sqrt{27}=7\sqrt{3}-54\sqrt{2}$

ក. $-2(2\sqrt{12}-\sqrt{18})-5(3\sqrt{32}-\sqrt{27})$

ខ. $3(3\sqrt[3]{40}-\sqrt[3]{135})+4(\sqrt[3]{320}-\sqrt[3]{40})$

គ. $\frac{2\sqrt{27}}{3}-3\sqrt{48}+\frac{4\sqrt{50}}{5}-\frac{4\sqrt{18}}{3}$

ឃ. $\frac{2}{3}\sqrt[3]{81}-\frac{1}{2}\sqrt[3]{24}+\frac{2\sqrt[3]{135}}{3}-\frac{3\sqrt[3]{40}}{2}$

13 គណនាកន្សោមខាងក្រោម $(a,b,x,y,z>0)$ ៖

➥ **តំរូ ៖** $A=-12\sqrt{2x}+12\sqrt{2x}=0$ $G=36a\sqrt[3]{2a}$

$A=-3\sqrt{32x}+6\sqrt{8x}$

$B=2\sqrt{125x^2z}+8x\sqrt{80z}$

$C=7a\sqrt{b^3}+b\sqrt{4a^2b}-\sqrt{4b}$

$D=8b\sqrt{49b}-7\sqrt{9b^3}+a\sqrt{4a}+\sqrt{a^3}$

$E=3xy\sqrt{x^2y}-2\sqrt{x^4y^3}$

$F=-3a\sqrt{a^3b^5}-2b\sqrt{a^5b^3}+5\sqrt{a^5b^3}$

$G=8a\sqrt[3]{54a}+6\sqrt[3]{16a^4}$

$H=3\sqrt[3]{x^4y}-6x\sqrt[3]{xy^4}+2\sqrt[3]{x^4y^4}$

14 គណនាតម្លៃលេខនៃកន្សោម ៖

➥ **តំរូ ៖** ① ជំនួស $a=5,b=3$ ៖ $A=\sqrt{15}-3\sqrt{15}-45\sqrt{15}-5\sqrt{15}=-52\sqrt{15}$

① $A=\sqrt{ab}-\sqrt{ab^3}-\sqrt{9a^3b^3}-\sqrt{a^3b}$ ចំពោះ $a=5,b=3$

② $A=\sqrt{4a}+a\sqrt{a^2b}+\sqrt{b^2a}+b\sqrt{9b}$ ចំពោះ $a=3,b=2$

15 គណនា ៖

➥ **តំរូ ៖** ក. $6\sqrt{6}$ ជ. $15\sqrt[3]{30}$ ញ. $3\sqrt{5}(6\sqrt{2}-12\sqrt{3})=18\sqrt{10}-36\sqrt{15}$

ក. $(2\sqrt{3})(3\sqrt{2})$ ខ. $(4\sqrt{6})(-2\sqrt{5})$

គ. $(3\sqrt{5})(5\sqrt{3})$ ឃ. $(6\sqrt{2})(-12\sqrt{3})$

ង. $(3\sqrt[3]{8})(-3\sqrt{48})$ ច. $(-3\sqrt{75})(-2\sqrt{48})$

ឆ. $(2\sqrt[3]{3})\left(-\frac{1}{2}\sqrt[3]{2}\right)$ ជ. $(3\sqrt[3]{2})(5\sqrt[3]{15})$

ឈ. $(6\sqrt[3]{8})(-3\sqrt[3]{2})$ ញ. $3\sqrt{5}(2\sqrt{18}-3\sqrt{48})$

ដ. $-3\sqrt{3}(3\sqrt{6}-3\sqrt{2})$ ឋ. $\frac{1}{2}\sqrt{3}(2\sqrt{48}-3\sqrt{32})$

ឌ. $\frac{3}{2}\sqrt{2}(2\sqrt{18}-3\sqrt{48})$ ឍ. $-4\sqrt[3]{3}(2\sqrt[3]{6}-2\sqrt[3]{5})$

ណ. $2\sqrt[3]{5}(3\sqrt[3]{3}-5\sqrt[3]{2})$

16 គណនា ៖

➥ **តំរូ ៖** ក. $18\sqrt{3}+4\sqrt{15}-72-16\sqrt{5}$ គ. $100-40\sqrt{6}$

ក. $(2\sqrt{3}-8)(9+2\sqrt{5})$

ខ. $(3\sqrt{5}-2\sqrt{10})(\sqrt{50}-2\sqrt{80})$

គ. $(\sqrt{50}-\sqrt{75})(\sqrt{32}-\sqrt{48})$

ឃ. $(\sqrt{125}-\sqrt{75})(\sqrt{80}-\sqrt{48})$

ង. $(3\sqrt[3]{18}+3\sqrt[3]{27})(2\sqrt[3]{8}-2\sqrt[3]{12})$

ច. $(3\sqrt[3]{80}-2\sqrt[3]{27})(-3\sqrt[3]{20}-3\sqrt[3]{12})$

17 គណនា ៖

➥ **គំរូ ៖** ក. $\frac{\sqrt{33}}{\sqrt{121}} = \sqrt{\frac{3}{11}}$ ច. $\frac{12\sqrt[3]{6}}{\sqrt[3]{2}} = 12\sqrt[3]{3}$

ក. $\frac{\sqrt{33}}{11}$ ខ. $\frac{-\sqrt{48}}{\sqrt{27}}$ គ. $2\sqrt{75} \div \sqrt{15}$

ឃ. $\frac{2\sqrt{84}}{-\sqrt{12}}$ ង. $\frac{5\sqrt{25}}{\sqrt{75}} \div \sqrt{5}$ ច. $\frac{12\sqrt[3]{6}}{\sqrt[3]{2}}$

ឆ. $\frac{5\sqrt[3]{100}}{25\sqrt[3]{5}}$ ជ. $\frac{\sqrt[3]{72}}{2\sqrt[3]{8}} \cdot \sqrt[3]{2}$ ឈ. $\frac{6\sqrt[3]{7}}{3\sqrt[3]{6}} : \frac{4\sqrt[3]{21}}{2\sqrt[3]{3}}$

18 គណនា ៖

ក. គេឱ្យ $a = 3\sqrt{5} - 2\sqrt{10}$, $b = 5\sqrt{7} + 2\sqrt{10}$, $c = \sqrt[3]{18} - \sqrt[3]{27}$ និង $d = 3\sqrt[3]{6} + \sqrt[3]{8}$ ៖

➥ **គំរូ ៖** ក. ❶ $-3ab = -3(3\sqrt{5} - 2\sqrt{10})(5\sqrt{7} + 2\sqrt{10})$

❶ $-3ab$ ❷ $a^2 + b^2$ ❸ $a^2 - 2b^2$

❹ $b^2 - 2ab$ ❺ $\frac{1}{2}cd$ ❻ $c^2 - b^2$

ខ. គេឱ្យ $m = 3\sqrt{8} + \sqrt{5}$ និង $n = 3\sqrt{8} - \sqrt{5}$ ៖

➥ **គំរូ ៖** ខ. ❶ $\frac{mn+m^2}{m} = \frac{m(n+m)}{m} = n + m = 12\sqrt{2}$

❶ $\frac{mn+m^2}{m}$ ❷ $\frac{m^2-n^2}{m+n}$ ❸ $\frac{n^2-2mn}{n^2}$

IV ទិធីបំបាត់រ៉ាឌីកាល់ពីភាគបែង

19 បំបាត់រ៉ាឌីកាល់ពីភាគបែង ៖

➥ **គំរូ ៖** ក. $\frac{3\sqrt{10}}{10}$ ខ. $\frac{\sqrt[3]{900}}{10}$ ជ. $\frac{\sqrt{3}+1}{2}$ ម. $\frac{\sqrt[3]{4}-\sqrt[3]{2}+1}{3}$

ក. $\frac{3}{\sqrt{10}}$ ខ. $\frac{3\sqrt{3}}{\sqrt{33}}$ គ. $\frac{6}{\sqrt{48}}$

ឃ. $\frac{8}{\sqrt{27}}$ ង. $\frac{9\sqrt{4}}{\sqrt{18}}$ ច. $\frac{5\sqrt{5}}{\sqrt{72}}$

ឆ. $\frac{4}{\sqrt{8}}$ ជ. $\frac{36-\sqrt{6}}{\sqrt{8}}$ ឈ. $\frac{\sqrt{3}+\sqrt{5}}{3\sqrt{20}}$

ញ. $\frac{15}{\sqrt{50}}$ ដ. $\frac{5-\sqrt{10}}{\sqrt{5}}$ ឋ. $\frac{6+2\sqrt{3}}{\sqrt{12}}$

ឌ. $\frac{\sqrt[3]{9}}{\sqrt[3]{10}}$ ឍ. $\frac{2\sqrt[3]{3}}{\sqrt[3]{30}}$ ណ. $\frac{\sqrt[3]{18}}{\sqrt[3]{20}}$

ត. $\frac{4}{\sqrt[3]{2}}$ ថ. $\frac{9-\sqrt[3]{3}}{2\sqrt[3]{32}}$ ទ. $\frac{5\sqrt[3]{4}+\sqrt[3]{3}}{8\sqrt[3]{13}}$

ធ. $\frac{1}{\sqrt{3}-1}$ ន. $\frac{4}{\sqrt{7}-\sqrt{3}}$ ប. $\frac{\sqrt{5}-1}{\sqrt{5}+1}$

ផ. $\frac{6}{\sqrt{11}-\sqrt{5}}$ ព. $\frac{8}{2\sqrt{75}-3\sqrt{50}}$ ភ. $\frac{2\sqrt{3}}{2\sqrt{80}-\sqrt{45}}$

ម. $\frac{1}{\sqrt[3]{2}+1}$ យ. $\frac{2}{\sqrt[3]{3}-1}$ រ. $\frac{3}{\sqrt[3]{4}+1}$

ល. $\frac{1}{\sqrt[3]{4}-1}$ វ. $\frac{1}{\sqrt[3]{5}-\sqrt[3]{2}}$ ស. $\frac{4}{\sqrt[3]{7}+\sqrt[3]{3}}$

20 បំបាត់រ៉ាឌីកាល់ពីភាគបែង $(x, y, a, b, m, n > 0)$ ៖

➥ **គំរូ ៖** ក. $\frac{3\sqrt{x}}{x}$ ច. $\frac{15(\sqrt{2a}+\sqrt{b})}{2a-b}$

ក. $\frac{3}{\sqrt{x}}$ ខ. $\frac{3a}{\sqrt{8ab}}$

គ. $\frac{3\sqrt{x}}{5\sqrt{2x^3y}}$ ឃ. $\frac{8a}{\sqrt[3]{25ab}}$

ង. $\frac{27\sqrt[3]{m}}{7\sqrt[3]{m^2n}}$ ច. $\frac{15}{\sqrt{2a}-\sqrt{b}}$

ឆ. $\frac{7\sqrt{2x}}{2\sqrt{x}+\sqrt{3y}}$ ជ. $\frac{\sqrt[3]{11}}{\sqrt[3]{2m}-\sqrt[3]{12n}}$

V ចំណោទអនុវត្ត និងលំហាត់ពាក្យ

21 គេឱ្យត្រីកោណកែងមួយមានរង្វាស់ជ្រុង a, b, c (c ជាអ៊ីប៉ូតេនុស) ៖

➥ **គំរូ ៖** ❶ $c = \sqrt{3^2 + 4^2} = 5$ ❹ $a = \sqrt{c^2 - b^2} = 24$

❶ $a = 3, b = 4$ ❷ $a = 5, b = 12$

❸ $a = 6, b = 8$ ❹ $c = 26, b = 10$

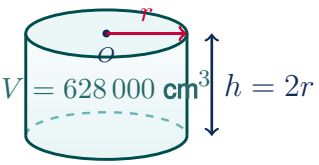
❺ $b = 30, c = 50$

22 គណនាកាំ r នៃស៊ីឡាំង ៖

គេមានស៊ីឡាំងមួយមានមាឌ $V = 628\,000\ \text{cm}^3$ និងកម្ពស់ $h = 2r$ (យក $\pi = 3.14$) ។ គណនាកាំ r នៃបាតស៊ីឡាំង ៖

➥ **គំរូ ៖** $V = \pi r^2 h = 2\pi r^3 \implies r = \sqrt[3]{\frac{V}{2\pi}}$

$r = \sqrt[3]{\frac{628\,000}{2 \times 3.14}} = \sqrt[3]{100\,000} = 100\ \text{cm}$



23 លំហាត់ពង្រឹងស្វ័យគុណ ៖

➥ **គំរូ ៖** ❶ $2^{17} + 2^{17} = 2(2^{17}) = 2^{18}$ ❺ $2^{2x+3} = 2^{15} \implies x = 6$

ក. បង្ហាញសមភាព ៖

❶ $2^{17} + 2^{17} = 2^{18}$ ❷ $3^{15} + 3^{15} + 3^{15} = 3^{16}$

❸ $4^8 + 4^8 = 2^{17}$ ❹ $9^5 + 9^5 + 9^5 = 3^{11}$

ខ. រកតម្លៃ x ៖

❺ $2^x \cdot 2^{x+3} = 8^5$ ❻ $3^x \cdot 3^{x+1} = 27^3$

❼ $\frac{2^{3x-1}}{2^{x+1}} = 16$ ❽ $4^{x+1} = 2^{3x-2}$

គ. គណនា និងប្រៀបធៀប ៖

❾ $A = \frac{2^{10} \cdot 3^8}{6^7}$ ❿ ប្រៀបធៀប 2^{30} និង 3^{20}